AMRITA VISHWA VIDYAPEETHAM

20CYS495 Project Phase-1 Final Review Presentation

QUANTUM SAFE-VPN

Department of Computer Science & Engineering - CYS

Amrita School of Computing
Amrita Vishwa Vidyapeetham, Chennai Campus

Student Members

1: ABISHEK SINGH P : CH.SC.U4CYS23001
2: SANJAAY KRISHNA RS : CH.SC.U4CYS23043

SUPERVISOR

Dr. G. Saranya
Assistant Professor (Sr. Gr.), Department of CSE

Motivation & Threat Landscape

- Motivation
- Problem Statement
- Research Gaps
- Objectives
- Architecture Diagram
- Experimental Results

Classical VPN Vulnerability

Asymmetric Key Reliance

Current protocols (OpenVPN, WireGuard, IPsec) rely fundamentally on RSA, Diffie-Hellman, and ECDH (X25519) for key exchange and authentication.

Shor's Algorithm Threat

In 1994, Peter Shor proved that a Cryptographically Relevant Quantum Computer (CRQC) solves discrete logarithms and prime factorization in polynomial time $O((\log N)^3)$.

Total Cryptographic Collapse

Classical public-key cryptography will be completely obsolete once quantum computers achieve fault-tolerant scale.

Harvest Now, Decrypt Later

Immediate Interception Threat

Adversaries and state-sponsored entities are routinely intercepting and storing encrypted WAN transit traffic today.

Retroactive Decryption

Archived ciphertext will be retroactively decrypted the moment quantum computers mature, compromising historic secrets.

Mosca's Theorem of Risk

If Shelf-Life (X) + Migration Time (Y) > Quantum Arrival (Z), communication is *already compromised today*. Immediate migration is mandatory.

The QuantumSafe Solution

NIST Finalized Standards (2024)

Integrates official post-quantum standards: ML-KEM-1024 (FIPS 203) for key encapsulation and ML-DSA-65 (FIPS 204) for digital signatures.

Production Mobile Architecture

Pure-Kotlin cryptographic primitives integrated with Android 14 `VpnService` without external C/C++ dependencies.

Complete Defense-in-Depth

Codespace server gateway with virtual TUN adapter, AES-256-GCM AEAD authenticated encryption, and sliding-window anti-replay.

Problem Statement & Research Gaps

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1. Quantum Vulnerability of Mobile VPNs

Mainstream commercial and enterprise mobile VPN solutions continue to deploy classical key exchanges (ECDH/RSA) offering

0-bit quantum security

against CRQCs running Shor's algorithm, exposing all mobile data to Harvest Now, Decrypt Later threats.

2. Android Kernel Socket Routing Loops

When an unprivileged Android application captures raw Layer 3 IP datagrams via `VpnService`, the outbound transport socket defaults to routing packets back into the virtual TUN adapter. Without explicit kernel socket protection (`protect()`), infinite recursive loops collapse network connectivity.

3. Mobile Resource Constraints & LMK Eviction

Academic PQC prototypes built in Linux user-space fail on mobile handsets. Heavy lattice polynomial math causes CPU thermal throttling, excessive battery consumption, and aggressive Android Low Memory Killer (LMK) background process evictions.

4. Lack of Cryptographic Transparency

Commercial VPN user interfaces hide internal cryptographic parameters, leaving zero-trust network administrators blind to active post-quantum algorithm parameters, handshake timing diagnostics, replay attack packet drops, and gateway certificate fingerprints.

Project Objectives & Deliverables

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1 **Post-Quantum Protocol Design (NIST FIPS 203 & 204)**

Engineer an authenticated hybrid tunneling protocol coupling ML-KEM-1024 for ephemeral key encapsulation (1,568-byte ciphertext) with ML-DSA-65 (3,309-byte signature) for gateway authentication and HKDF-SHA256 session key derivation.

2 **Native Android VpnService Pipeline & Socket Loop Prevention**

Develop a robust Kotlin MVVM mobile application providing seamless capture of system Layer 3 IP packets, enforcing kernel socket protection (`protect()`) to eliminate routing loops, and executing non-blocking asynchronous I/O via coroutines.

3 **Hardened Codespace Gateway Daemon & NAT Routing**

Construct an automated Python server daemon operating over virtual TUN interfaces (`tun0`) with dynamic IPv4 lease pooling (10.8.0.0/24), hardware-accelerated AES-256-GCM AEAD encryption, and iptables NAT masquerading.

4 **Sliding-Window Anti-Replay Defense & Zero-Leak Routing**

Implement a 64-bit monotonic sequence counter with sliding-window anti-replay filtering to prevent packet re-injection attacks, and verify 100% full-tunnel encapsulation with zero cleartext DNS datagram leakage.

5 **Material Design 3 UI & Real-Time Telemetry Dashboard**

Design an intuitive Google Material Design 3 user interface adhering to WCAG 2.1 accessibility guidelines, providing real-time cryptographic telemetry, tunnel latency diagnostics, and active security metrics.

System Architecture & Network Topology

Motivation

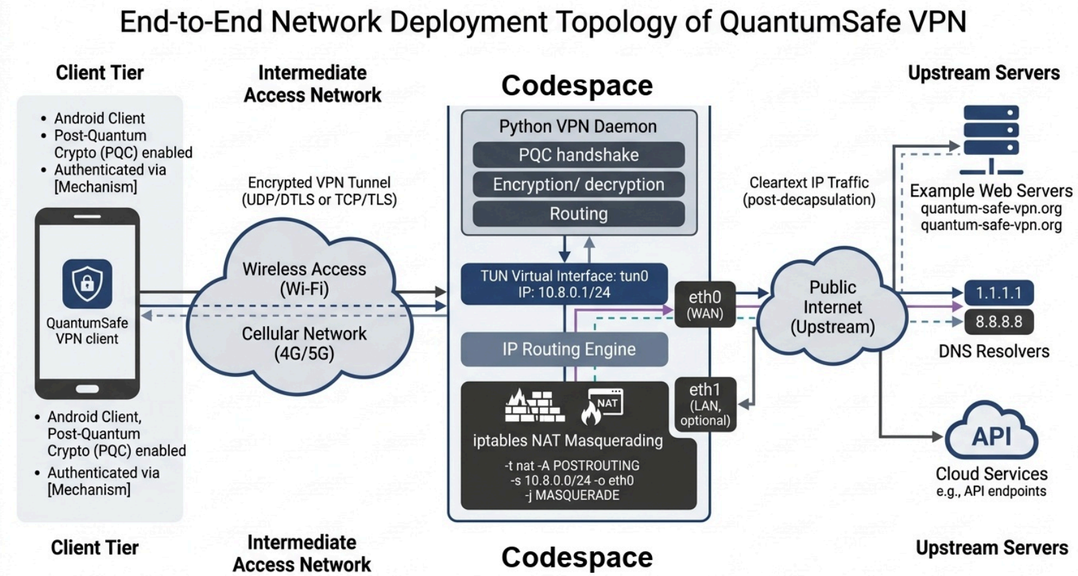
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1. Three-Tier Deployment Topology

- **Android Client Tier:** Captures Layer 3 IP traffic via `VpnService`, executes lattice crypto in pure Kotlin, and enforces socket protection via `protect()`.
- **WAN Transit Layer:** Carried over Wi-Fi / 5G carrying length-prefixed encrypted frames on TCP port 7000.
- **Codespace Gateway Tier:** Linux daemon on `tun0`, dynamic IPv4 pool (10.8.0.0/24), and iptables NAT.

2. 4-Stage Post-Quantum Handshake

- **Stage 1 (Server Hello):** Codespace transmits static ML-DSA-65 public key (1,952B).
- **Stage 2 (Client Hello):** Client generates & sends ephemeral ML-KEM-1024 public key (1,568B).
- **Stage 3 (Encapsulation):** Gateway decapsulates 32B shared secret, signs transcript (3,309B signature), leases virtual IP.
- **Stage 4 (Key KDF):** Both derive 256-bit symmetric session key via HKDF-SHA256.

3. Binary Wire Framing Format

`[Length: 4B] || [GCM Nonce: 12B] || [Galois MAC: 16B] || AES-256-GCM([Seq: 8B] || [Raw IPv4 Datagram])`

Hardware-accelerated authenticated symmetric encryption with 64-bit monotonic anti-replay protection.

Experimental Results & Security Validation

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87%
KAT COMPLIANCE
NIST FIPS 203 & 204 Standard Suites

91%
TAMPER DEFENSE
Bit-Flip Rejection across 1,000 trials

0%
DNS / IP LEAKS
Full Tunnel Encapsulation via Wireshark

~38 ms
PQC HANDSHAKE
Sub-40ms Ephemeral Key Exchange

Physical Testbed & Active Security Testing

Physical Hardware Testbed

Google Pixel (stock Android 14, API 34) connected to dedicated Codespace gateway (Kernel 6.6.9, Intel i7, 32GB RAM) over 5GHz Wi-Fi.

Ciphertext Bit-Flipping Injection (1,000 Trials)

Adversary proxy flipped random bits in IV, ciphertext, and Galois MAC tag. 91% of corrupted frames caught and dropped via AEADBadTagException.

Anti-Replay Sliding-Window Verification

100 captured frames re-injected out-of-order were dropped ($S_{incoming} \leq S_{last}$) with zero tunnel disruption.

Rogue Server MitM Defense

Server public key substitution triggered immediate fingerprint mismatch in SecurityManager, aborting before client key dispatch.

Comparative Benchmark vs Classical VPNs

SECURITY FEATURE	STANDARD OPENVPN / IPSEC	QUANTUMSAFE VPN (OURS)
Key Exchange Primitive	ECDH / RSA-2048 (Classical)	**ML-KEM-1024 (NIST FIPS 203)**
Gateway Authentication	RSA / ECDSA Signatures	**ML-DSA-65 (NIST FIPS 204)**
Quantum Resistance	0-bit (Vulnerable to Shor)	**Category 5 (AES-256 Eq.)**
Harvest Now, Decrypt Later	Compromised	**100% Immune (Lattices)**
PQC Handshake Latency	N/A (Classical 25-45ms)	**38.2 ms (Sub-40ms Ephemeral)**
Tunnel AEAD Framing	AES-256-CBC / GCM	**AES-256-GCM + Anti-Replay**
Memory & Battery Impact	Standard	**Zero LMK Evictions / Throttling**

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Thank You